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Differences in reproductive and productive performance and removal from the herd associated with time to clinical cure in dairy cows treated for metritis

F. N. S. Pereira,[1] E. B. de Oliveira,[2] V. R. Merenda,[3] F. S. Lima,[2] R. C. Chebel,[4,5] K. N. Galvão,[4] J. E. P. Santos,[5] R. S. Bisinotto,[4] and C. C. Figueiredo[1]*

[1]Department of Veterinary Clinical Sciences, Washington State University, Pullman, WA 99164
[2]Department of Population Health and Reproduction, University of California, Davis, CA 95616
[3]Department of Population Health and Pathobiology, North Carolina State University, Raleigh, NC 27606
[4]Department of Large Animal Clinical Sciences, D. H. Barron Reproductive and Perinatal Biology Research Program, University of Florida, Gainesville, FL 32610
[5]Department of Animal Sciences, University of Florida, Gainesville, FL 32608

ABSTRACT

The objectives of this study were to assess differences in lactational performance (reproduction, milk production, and removal from the herd) associated with time to clinical cure of metritis. This retrospective cohort study was conducted using data from 3 experiments conducted at 5 dairy herds in Florida. Cows with metritis were randomly assigned to receive either ampicillin or ceftiofur in 2 experiments (1 experiment only implemented ceftiofur). Vaginal discharge assessment using the Metricheck device within 12 DIM was used to diagnose metritis, characterized by presence of fetid, watery, reddish-brown vaginal discharge (VD; n = 1,410), and the day of diagnosis was defined as d 0. A total of 556 cows were treated with ampicillin, and 854 cows were treated with ceftiofur. Clinical cure was visually assessed on d 5 and 12 and defined as clear, mucopurulent, or purulent VD. Cows were grouped as follows: CureD5 (cured by d 5; n = 705); CureD12 (cured between d 6 and 12; n = 406); NoCure (not cured by d 12; n = 299); NoMet (not diagnosed with metritis; n = 1,194). Data regarding risk of receiving a first service, risk and hazard of pregnancy by 300 DIM, risk and hazard of removal from the herd by 300 DIM, and total milk production within 10 mo were collected from the herd management software. Statistical analyses were performed using multivariable logistic regression, Cox proportional hazards, and ANOVA models. Orthogonal contrasts were set to assess the effects of metritis, clinical cure, and time to clinical cure. The antimicrobial treatment formulation was not associated with the odds of cure by d 5. A greater proportion of CureD5 cows received a first service (91.6% vs. 87.1%) and were pregnant by 300 DIM (69.6% vs. 62.9%) compared with CureD12 cows. However, time to cure was not associated with reduced hazard of pregnancy within 300 DIM. A smaller proportion of CureD5 cows were removed from the herd by 300 DIM compared with CureD12 cows (15.5% vs. 23.7%). Additionally, a smaller hazard of removal from the herd within 300 DIM was observed for CureD5 compared with CureD12 cows (0.61). No differences in milk production within 10 mo were associated with time to clinical cure. Our findings highlight that expedited clinical cure of metritis was associated with positive effects on reproductive performance and removal from the herd.

Key words: uterine diseases, antimicrobials, cattle

INTRODUCTION

Metritis is a polymicrobial uterine disease that affects between 15 and 40% of postpartum dairy cows in the US (Pinedo et al., 2020; Menta et al., 2024) and across the world (Giuliodori et al., 2013; Pohl et al., 2016; Bruinjé et al., 2023). Metritis has been consistently associated with negative effects on reproduction (Ribeiro et al., 2013; Bruinjé et al., 2024a), milk production (de Oliveira et al., 2020; Figueiredo et al., 2024a), removal from the herd (Wittrock et al., 2011; Figueiredo et al., 2024a), and animal welfare (Stojkov et al., 2015; Barragan et al., 2018). Together, the detrimental effects of metritis contribute to an estimated annual economic loss of $1.2 billion to the US dairy industry and up to $5 billion worldwide (Pérez-Báez et al., 2021; Rasmussen et al., 2024).

The primary therapeutic approach for metritis is based on antimicrobials, and country-wide data indicate that metritis is the second most common reason for antimicrobial use in dairy cattle (USDA, 2018). Indeed, almost

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*Corresponding author: caio.figueiredo@wsu.edu

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60% of metritis cases are treated using antimicrobials (USDA, 2018), and recently published economic analyses have reinforced the importance of antimicrobial treatment for cows with metritis (Silva et al., 2021). de Oliveira et al. (2020) reported that cows with metritis that received antimicrobial treatment had improved fertility and reduced frequency of removal from the herd compared with nontreated cows with metritis. Despite the higher treatment costs associated with antimicrobial use, treatment was associated with a $250 reduction in cost per case of metritis compared with nontreated cows (Silva et al., 2021). In the United States, ceftiofur and oxytetracycline are antimicrobial formulations labeled for the treatment of metritis, and at least 75% of dairies use ceftiofur as the first treatment option (Espadamala et al., 2018; Juliano et al., 2025). Although ampicillin and penicillin are not labeled for the treatment of metritis, ~25% of dairy operations use one of these drugs as their first-line treatment (Espadamala et al., 2018). This practice is largely driven by producers' perception that cows treated with ampicillin or penicillin recover more quickly than those treated with ceftiofur (Espadamala et al., 2018). However, scientific studies evaluating the time to clinical cure across different antimicrobial therapies have yielded conflicting results. Lima et al. (2014) reported that a greater proportion of ampicillin-treated cows with metritis achieved clinical cure by d 7 after diagnosis compared with ceftiofur-treated cows with metritis (cows treated at diagnosis; 50.4% vs. 37.9%, respectively). Conversely, Merenda et al. (2021) reported a similar proportion of clinical cure between ampicillin- and ceftiofur-treated cows with metritis in the same period (33.9% vs. 37.5%, respectively). Considering that both antimicrobial formulations are β-lactams and that the uterine microbiome composition is highly similar between cured and noncured cows with metritis (Jeon et al., 2018; Figueiredo et al., 2024b), the relationship between antimicrobial formulation and the underlying mechanisms associated with time to clinical cure remains unclear.

Regardless of the inconsistent association between antimicrobial formulation and time to clinical cure of metritis, the literature regarding the differences in lactational performance associated with time to cure of metritis is scarce. One study that assessed time to cure of metritis (d 5 vs. d 14 after diagnosis) reported no differences in milk production, reproductive performance, or removal from the herd between groups (Menta et al., 2024). More studies are needed to understand whether cows that recover more rapidly from metritis outperform those that take longer to recover. This is particularly important because time to clinical cure has been a factor influencing antimicrobial use decisions and has contributed to extralabel use of different antimicrobial formulations (Espadamala et al., 2018). The objective of the present study was to evaluate the association between time to clinical cure of metritis and reproductive performance, milk production, and removal from the herd. We hypothesized that cows with expedited clinical cure would demonstrate better lactational performance than those with delayed cure.

MATERIALS AND METHODS

No animals were used specifically for this study; the original experiments from which the data were obtained received IACUC approval from the University of Florida Institutional Animal Care and Use Committee.

Study Population and Housing

This retrospective cohort study was generated using data from 3 experiments conducted at 5 dairy herds in Florida between 2012 and 2018 (herds 1 and 2: de Oliveira et al., 2020; herd 3: Lima et al., 2014; herds 4 and 5: Merenda et al., 2021). The number of lactating cows and rolling herd average for each herd follow: herd 1 - 4,400 and 12,500 kg; herd 2 - 1,800 and 10,500 kg; herd 3 - 5,270 and 11,000 kg; herd 4 - 2,300 and 10,333 kg; and herd 5 - 2,500 and 12,049 kg. Holstein cows were milked 2 (herd 4) or 3 times (herds 1, 2, 3, and 5) daily, in parallel parlors. Cows were fed a TMR to meet or exceed the nutritional requirements for a 650 to 680 kg Holstein cow producing 40 to 45 kg/d of 3.5% FCM (NRC, 2001). Cows were housed in freestall barns with sand bedding equipped with sprinklers and fans over the stalls and feeding alley. Additionally, the cows had ad libitum access to water. Cows were classified as primiparous (lactation = 1) or multiparous (lactation >1). Additional information regarding the dairies is provided in the previously published studies (Lima et al., 2014; de Oliveira et al., 2020; Merenda et al., 2021).

Calving was monitored by trained farm personnel, and cows were assisted as defined by each herd's standard operational procedures. Date and season of calving (spring = March–May; summer = June–August; fall = September–November; winter = December–February) were recorded for each cow. Calving difficulty was scored on a 4-point scale validated for dairy cattle: 1 = unassisted parturition; 2 = minimal assistance provided by 1 technician without the use of mechanical traction; 3 = mechanical extraction of the calf; 4 = severe dystocia requiring cesarean section or fetotomy (Schuenemann et al., 2011). Dystocia was characterized by calving difficulty ≥2. Stillbirth was defined as a calf delivered dead or that died within 24 h after delivery. Calf sex and twinning were recorded by farm personnel. Retained fetal membranes (**RFM**) were characterized as membranes not detached by 24 h after parturition, based on visual inspection.

Definition of Metritis, Treatment, Clinical Cure

Metritis was diagnosed by a research team based on a visual evaluation of vaginal discharge (**VD**) using a Metricheck device (Simcro, Hamilton, New Zealand). The VD of each cow was evaluated at 4, 6, and 8 DIM (Lima et al., 2014); 5, 7, and 9 DIM (de Oliveira et al., 2020); or 4, 6, 8, 10, and 12 DIM (Merenda et al., 2021). Rectal temperature (**RT**) was monitored in all cows according to the protocols of each respective experiment in addition to the scheduled postpartum evaluations. Any febrile cow (RT ≥ 39.5°C) underwent an immediate evaluation of VD on the same day. Vaginal discharge was scored on a 5-point scale: 1 = clear mucus or lochia; 2 = clear mucus with flecks of pus; 3 = mucopurulent discharge with <50% of pus; 4 = mucopurulent discharge with ≥50% of pus or nonfetid reddish mucous discharge; 5 = fetid, watery, reddish-brown discharge (Figueiredo et al., 2024a). Metritis was diagnosed by the presence of VD score of 5 (n = 1,410). The day of metritis diagnosis was considered study d 0. Rectal temperature was recorded alongside the evaluation of VD. Cows with an RT ≥39.5°C were considered febrile (metritis with fever: n = 532; metritis without fever: n = 878). Cows without metritis (**NoMet**; VD score ≤4 between 4 and 12 DIM; n = 1,194) of similar parity and calving date were included in the study as a negative control to assess the association between metritis and relevant outcomes of interest. Body condition score was assessed on d 0 on a scoring scale of 1 (a very thin cow) to 5 (an excessively fat cow) with 0.25-unit increments (Ferguson et al., 1994). Cows were classified as having low (BCS ≤2.75), moderate (BCS 3.00–3.50), or high (BCS ≥3.75) body condition.

On d 0, cows diagnosed with metritis were randomly assigned to receive antimicrobial treatment in accordance with the experimental protocols of each contributing study. Antimicrobial treatments were applied across all 3 experiments, including 11 mg/kg of BW of ampicillin trihydrate i.m. (Polyflex, Boehringer Ingelheim Vetmedica, Duluth, GA) injected once daily for 5 consecutive days (n = 556; Lima et al., 2014; Merenda et al., 2021); 2.2 mg/kg of BW of ceftiofur hydrochloride i.m. (Excenel RTU Sterile Suspension, Zoetis, Madison, NJ) injected once daily for 5 consecutive days (n = 302; Lima et al., 2014); or ceftiofur crystalline free acid s.c. (Excede Sterile Suspension, Zoetis, Madison, NJ) injected twice, 72 h apart (n = 552; de Oliveira et al., 2020; Merenda et al., 2021). Upon diagnosis of metritis, the approximate BW of each cow was determined with a heart-girth tape (Weight-By-Breed Dairy Management Tape, Nasco, Fort Atkinson, WI) to calculate the volume of the antimicrobial to be administered.

Clinical cure was assessed based on the evaluation of VD characteristics in all cows with metritis on d 5 and 12, as previously described. Clinical cure was defined as cows with VD score ≤4, whereas clinical cure failure was defined as cows still displaying VD score of 5 on d 5 or 12. The VD of cows that were not clinically cured by d 5 was reassessed on d 12. Cows were grouped according to diagnosis of metritis on d 0 and time to clinical cure, as follows: cows diagnosed with metritis that achieved clinical cure by d 5 (**CureD5**; n = 705); cows diagnosed with metritis that achieved clinical cure between d 6 and 12 (**CureD12**; n = 406); cows diagnosed with metritis that did not achieve clinical cure by d 12 (**NoCure**; n = 299); and NoMet cows (n = 1,194).

Evaluation of Purulent VD

Occurrence of purulent VD (**PVD**) was evaluated in a subgroup of cows (NoMet: n = 910 [76.2% of total NoMet population]; CureD5: n = 554 [78.6% of total CureD5 population]; CureD12: n = 340 [83.7% of total CureD12 population]; NoCure: n = 227 [75.9% of total NoCure population]). Vaginal discharge was assessed at 30 ± 5 DIM with the Metricheck device, and PVD was characterized by the presence of a VD score of ≥3.

Reproductive Management and Reproductive Performance

In herd 1, cows underwent a 48-d voluntary waiting period (**VWP**) from the calving date before being observed daily for estrus signs with a mount detection patch (Estrotect, Rockway). If cows were not inseminated by 55 ± 3 DIM, they received a $PGF_{2\alpha}$ treatment. Those still not inseminated by 72 ± 3 DIM were placed on a 7-d Ovsynch-56 protocol: an initial GnRH injection, followed by $PGF_{2\alpha}$ 7 d later, another GnRH injection 56 h after that, and timed artificial insemination (**AI**) 16 h later. Cows were checked for pregnancy 33 ± 3 d after AI by means of transrectal ultrasonography and again at 70 ± 3 d after the AI via rectal palpation. Nonpregnant cows received a GnRH injection and were re-enrolled in the 7-d Ovsynch-56 protocol 7 d later. In herd 2, cows were observed daily for estrus signs after a 50-d VWP based on removal of tail chalk. Those not inseminated by 70 ± 3 DIM were enrolled in a 7-d Ovsynch-56 protocol. Pregnancy was diagnosed via rectal palpation at 43 ± 3 and 70 ± 3 post-AI; nonpregnant cows were re-enrolled in the same protocol. In herd 3, after a 60-d VWP, cows were observed daily for estrus signs based on the amount of tail chalk. Cows received an injection of $PGF_{2\alpha}$ at 50 ± 3 and 64 ± 3 DIM, and those that were not successfully inseminated by 76 ± 3 DIM were enrolled in a 5-d Cosynch-72 protocol: GnRH, followed by $PGF_{2\alpha}$ 5 and 6 d later, followed by GnRH and timed AI 2 d later. Pregnancy was diagnosed via transrectal ultrasonography 34 ± 3 and 62 ± 3 d after AI, and nonpregnant cows were enrolled in the 5-d Cosynch-72 protocol. In herds 4 and 5, cows received injections of $PGF_{2\alpha}$ (herd 4: 35 ± 3 and 49 ± 3 DIM; herd 5: 39 ± 3 and 53 ± 3 DIM) and were enrolled in a 7-d Cosynch-72 protocol—GnRH, followed by $PGF_{2\alpha}$ 7 d later, followed by GnRH and timed AI 3 d later—starting at 63 ± 3 DIM (i.e., no estrus detection before first AI, VWP = 63 d). Pregnancy was checked via rectal palpation 40 ± 3 d after AI and transrectal ultrasonography 70 ± 3 d after AI; nonpregnant cows were enrolled in the 7-d Cosynch-72 protocol. Reproductive outcomes consisted of the proportion of pregnant cows by 300 DIM and time to pregnancy within 300 DIM, defined as DIM at AI that resulted in a pregnancy confirmed at the second pregnancy examination. Because loss of data to follow-up, such as cows marked as do-not-breed or culled before reaching VWP, can introduce bias into the analysis of specific reproductive outcomes, the analysis included the proportion of cows that received a first service during lactation to improve interpretation.

Milk Production and Removal from the Herd

Milk production was recorded through the monthly DHIA test in herds 1, 2, 4, and 5 on the herd's management software (milk data were not available for herd 3). Monthly milk production for the first 10 mo and total milk production for the first 10 mo postpartum from CureD5 (n = 426), CureD12 (n = 215), NoCure (n = 212), and NoMet (n = 853) data were collected. Date and DIM of removal from the herd and reason for removal (i.e., died or culled) were recorded for cows that left the herd within 300 DIM.

Sample Size Calculations and Statistical Analyses

This retrospective cohort study combined data from 3 previously published experiments that assessed clinical cure following antimicrobial therapy for metritis in dairy cows, along with subsequent fertility and production outcomes (Lima et al., 2014; de Oliveira et al., 2020; Merenda et al., 2021). Thus, sample size and power calculations were not performed for this study, as the dataset used herein consisted of a predetermined number of cows previously enrolled in the aforementioned experiments. Nevertheless, using α = 0.05 and β = 0.20, Lima et al. (2014) enrolled a minimum of 259 with metritis per antimicrobial group to detect a 9-percentage unit difference in cure on d 7 (e.g., 41 vs. 32 or 50%), or 7-percentage unit difference in cure on d 12 (77 vs. 69 or 84%) between ceftiofur and ampicillin. Merenda et al. (2021) enrolled a minimum of 304 cows with metritis per antimicrobial group to detect a 7-percentage unit difference in cure on d 11 after diagnosis (e.g., ceftiofur = 75% vs. ampicillin = 68%). Finally, de Oliveira et al. (2020) enrolled a minimum of 275 cows with metritis per treatment group (ceftiofur vs. no treatment) to detect treatment differences in metritis cure of 10-percentage units.

Univariable models were primarily performed to assess associations between independent variables and outcomes of interest. Variables with $P \leq 0.10$ in univariable analyses were considered for inclusion in the multivariable models. Variables with $P > 0.10$ were excluded by means of a backward stepwise elimination approach, in which variables were sequentially removed from the highest to the lowest P-value. The variables study (i.e., Lima et al., 2014; de Oliveira et al., 2020; Merenda et al., 2021) and group (i.e., NoMet, CureD5, CureD12, and NoCure) were forced into all final models regardless of their significance. Binary outcomes were analyzed by logistic regression in SAS with the LOGISTIC and FREQ procedures in SAS. Initial models evaluating risk factors associated with clinical cure on d 5 in cows (only CureD5 and CureD12 were included in this analysis) diagnosed with metritis included the fixed effects of study, parity (primiparous or multiparous), calving season (fall, winter, spring, or summer), calving problem (yes or no), RFM (yes or no), BCS category (low, moderate, or high BCS), fever (yes or no), and DIM at metritis diagnosis. For statistical purposes, a "calving problem" was defined as the occurrence of at least one of the following: dystocia, stillbirth, or twinning. Results were reported as odds ratios (**OR**) with 95% CI for both univariable and multivariable models. Additional logistic regression models were used to analyze PVD, risk of receiving a first service, pregnancy by 300 DIM, and removal from the herd by 300 DIM using the FREQ and GLIMMIX procedures of SAS. To acquire adjusted proportions from logistic regression models, the term LSMEANS within GLIMMIX was used. Initial models for these outcomes included the fixed effects of study, parity, group, calving season, calving problem, RFM, and BCS category. Multicollinearity among explanatory variables was assessed using the REG procedure with the variance inflation factor (**VIF**) and COLLIN options. Collinearity was defined as VIF >2. Cox proportional hazards models were performed with the PHREG procedure in SAS to assess the effects of clinical cure on d 5 on time to pregnancy and time to removal from the herd. Cox regression assumption was visually assessed via log-log survival curves. Kaplan-Meier survival curves were generated from the LIFETEST procedure. For analyses of time to pregnancy, cows were right-censored if they were not pregnant by the date of removal from the herd or by 300 DIM. For analyses of time to removal from the herd, cows were right-censored if they remained in the herd beyond 300 DIM. Initial models included the fixed effects of study, parity, group, calving season, calving problem, RFM, and BCS category. Results were reported as hazard ratios with 95% CI. Milk production for the first 10 mo postpartum was analyzed by ANOVA for repeated measures using the GLIMMIX procedure fitting a normal distribution and considering cow nested within cure group as a random variable. Initial models included the fixed effects of farm, group, parity, month postpartum, and relevant 2-way (group and month postpartum) and 3-way interactions (group, month postpartum, and parity). Heterogeneous first-order autoregressive was selected as structure of covariance because it resulted in the smallest Bayesian and Akaike's information criteria. Total milk production within 10 mo of lactation was analyzed by ANOVA using the GLIMMIX procedure, fitting a normal distribution. Fixed effects in the initial models were study, group, and parity. Assumptions of normality of residuals and homogeneity of variance were evaluated with the UNIVARIATE procedure.

In all analyses, orthogonal contrasts were set to assess the effects of metritis (CureD5 + CureD12 + NoCure vs. NoMet); cure (CureD5 + CureD12 vs. NoCure); and time to cure (CureD5 vs. CureD12). The Tukey–Kramer method was used to adjust for multiple comparisons. Differences were considered significant at $P \leq 0.05$, and tendencies were declared at $0.05 < P \leq 0.10$.

RESULTS

Descriptive Statistics

Incidences of metritis reported in each study were 36.1% (Lima et al., 2014), 25.3% (de Oliveira et al., 2020), and 19.0% (Merenda et al., 2021). Descriptive statistics regarding number of cows enrolled per group, parity, average lactation, calving problems (dystocia, stillbirth, twining), RFM, average DIM, BCS on day of enrollment, fever on day of enrollment, and the number of cows treated with ampicillin and ceftiofur are presented in Table 1. Overall, the proportion of multiparous cows was numerically higher in all groups, except for CureD12. Calving problems and RFM were more frequent in cows with metritis, particularly in the NoCure group, followed by CureD12 and CureD5. Cows were enrolled earlier postpartum in CureD12 and NoCure compared with CureD5 and NoMet. The proportion of cows with fever on the day of diagnosis was highest in NoCure, followed by CureD12 and CureD5. Across treatment groups, ceftiofur was administered to a greater proportion of cows than ampicillin.

Risk Factors Associated with CureD5

Parity, RFM, fever on d 0, and DIM at metritis diagnosis were associated with odds of CureD5 ($P \leq 0.03$; Table 2). Multiparous cows had greater odds of CureD5 after diagnosis compared with primiparous cows (OR: 1.71; 95% CI: 1.31–2.24), cows with RFM had lower odds of CureD5 after diagnosis compared with cows not

Table 1. Descriptive statistics across study groups for population, parity, lactation, calving problems, RFM, DIM, BCS, fever, and antimicrobial treatment[1]

Item	NoMet	CureD5	CureD12	NoCure
Total cows, n (%)	1,194	705	406	299
Primiparous cows, n (%)	418 (35.0)	295 (41.8)	216 (53.2)	142 (47.5)
Multiparous cows, n (%)	776 (65.0)	410 (58.2)	190 (46.8)	157 (52.5)
Average lactation (SD)	2.31 (1.3)	2.17 (1.3)	2.02 (1.4)	2.16 (1.5)
Calving problem,[2] n (%)	170 (14.2)	169 (24.0)	114 (28.1)	100 (33.4)
RFM,[3] n (%)	75 (6.3)	120 (17.0)	111 (27.3)	111 (37.1)
Average DIM on d 0[4] (SD)	7.26 (3.6)	7.26 (2.6)	5.80 (1.9)	5.38 (2.2)
Average BCS on d 0 (SD)	3.33 (0.3)	3.12 (0.4)	3.07 (0.3)	3.10 (0.3)
Fever on d 0, n (%)	—	206 (29.2)	184 (45.3)	142 (47.5)
Received ampicillin, n (%)	0 (0)	278 (39.4)	156 (38.4)	122 (40.8)
Received ceftiofur, n (%)	0 (0)	427 (60.6)	250 (61.6)	177 (59.2)

[1]NoMet: cows not diagnosed with metritis; CureD5: cows diagnosed with metritis within 12 DIM that achieved clinical cure by 5 d after diagnosis and treatment; CureD12 = cows diagnosed with metritis within 12 DIM that achieved clinical cure between 6 and 12 d after diagnosis and treatment; NoCure = cows diagnosed with metritis within 12 DIM that did not achieve clinical cure by 12 d after diagnosis and treatment. Data reported as LSM ± SEM and proportions are reflected within columns.

[2]Calving problem: cow that experienced dystocia, twins, or stillbirth.

[3]RFM: retained fetal membranes.

[4]d 0: day of metritis diagnosis.

Table 2. Risk factors associated with clinical cure by d 5 (after diagnosis) in cows treated for metritis

		Univariable analyses[1]		Multivariable analyses[1]	
Risk factor	Cure on D5, % (n)	OR[2] (95% CI)	*P*-value	OR[2] (95% CI)	*P*-value[3]
Study					
Lima et al. (2014)	27.9 (54)	Referent	<0.01	Referent	0.13
de Oliveira et al. (2020)	40.7 (191)	1.77 (1.23–2.55)		1.20 (0.73–1.97)	
Merenda et al. (2021)	35.9 (161)	1.23 (0.94–1.61)		0.72 (0.47–1.10)	
Parity					
Primiparous	57.7 (295)	Referent		Referent	
Multiparous	68.3 (410)	1.58 (1.24–2.02)	<0.01	1.71 (1.31–2.24)	<0.01
Calving season					
Spring	68.2 (161)	Referent	0.04	Referent	0.18
Summer	70.3 (83)	1.11 (0.68–1.79)		1.43 (0.85–2.40)	
Fall	63.2 (228)	0.80 (0.56–1.13)		1.06 (0.64–1.77)	
Winter	58.8 (233)	0.66 (0.47–0.94)		0.81 (0.55–1.21)	
Calving problem					
No	64.7 (536)	Referent	0.13	—	—
Yes	59.7 (169)	0.81 (0.61–1.07)		—	
RFM					
No	66.5 (585)	Referent	<0.01	Referent	<0.01
Yes	51.9 (120)	0.55 (0.41–0.73)		0.59 (0.42–0.82)	
BCS on d 0					
Low	59.3 (178)	0.79 (0.60–1.04)		—	
Moderate	64.9 (472)	Referent	0.22	—	—
High	65.5 (55)	1.03 (0.64–1.65)		—	
Fever on d 0					
No	69.2 (499)	Referent	<0.01	Referent	<0.01
Yes	52.8 (206)	0.50 (0.39–0.64)		0.57 (0.43–0.75)	
DIM on d 0 (per 1-d increase)	—	1.33 (1.25–1.41)	<0.01	1.31 (1.22–1.40)	<0.01

[1]A total of 1,111 cows were used for analysis, which consisted of 705 cows that cured by d 5, and 406 cows that cured between d 6 and 12.

[2]OR: odds ratio from univariable and multivariable logistic regression models.

[3]Nonsignificant variables ($P > 0.10$) are described in multivariable analyses for interpretation but were removed from final models.

affected by RFM (OR: 0.59; 95% CI: 0.42–0.82), and febrile cows on d 0 had lower odds of CureD5 after diagnosis compared with nonfebrile cows (OR: 0.57; 95% CI: 0.43–0.75). Each additional DIM at metritis diagnosis was associated with increased odds of CureD5 (OR: 1.31; 95% CI: 1.22–1.40). Conversely, study, calving season, calving problems, and BCS on d 0 were not ($P \geq 0.13$; Table 2) associated with different odds of CureD5.

Risk of PVD and Reproductive Performance

The prevalence of PVD was associated with metritis, clinical cure, and time to cure ($P < 0.01$; Table 3). The proportion of cows with PVD was 54.2% ± 0.04% in NoMet, 80.0% ± 0.02% in CureD5, 89.7% ± 0.02% in CureD12, and 94.8% ± 0.01% in NoCure. Cows with metritis had a greater proportion of PVD than those without metritis. Among cows with metritis, those that achieved clinical cure had a smaller proportion of PVD than those that did not achieve clinical cure. Additionally, cows in the CureD5 group had a lower proportion of PVD than those in the CureD12 group. Differences in proportion of PVD were associated with parity (primiparous = 86.8% vs. multiparous = 80.6%; $P < 0.01$), calving problems (No = 81.8% vs. Yes = 85.9%; $P = 0.03$), and RFM (No = 75.9% vs. Yes = 89.5%; $P < 0.01$).

Different proportions of cows receiving a first service were associated with metritis, clinical cure, and time to cure ($P \leq 0.01$; Table 3). The proportion of cows that received a first service was 92.6% ± 0.01% in NoMet, 91.6% ± 0.01% in CureD5, 87.1% ± 0.02% in CureD12, and 83.2% ± 0.02% in NoCure. Overall, a smaller proportion of cows with metritis received first service compared with those without metritis. A higher proportion of cows that achieved clinical cure received first service compared with those that did not achieve clinical cure. A higher proportion of CuredD5 cows received first service than CureD12 cows. In addition, a higher proportion of primiparous cows received first service compared with multiparous cows (92.2% vs. 85.1%; $P < 0.01$).

The proportion of pregnant cows by 300 DIM differed among the study groups ($P < 0.01$; Table 3). The proportion of pregnant cows by 300 DIM was 73.2% ± 0.02% in NoMet, 69.6% ± 0.02% in CureD5, 62.9% ± 0.03% in CureD12, and 56.4% ± 0.03% in NoCure. A greater proportion of cows without metritis achieved pregnancy

Table 3. Effect of metritis status, clinical cure and time to cure on purulent vaginal discharge (PVD), reproductive performance, removal from the herd, and milk production

Item	Group[1]				P-value[2]			
	NoMet	CureD5	CureD12	NoCure	Group	Metritis	Cure	Time
PVD, %	54.2 ± 0.04[a]	80.0 ± 0.02[b]	89.7 ± 0.02[c]	94.8 ± 0.01[d]	<0.01	<0.01	<0.01	<0.01
Received a first service, %	92.6 ± 0.01[a]	91.6 ± 0.01[ab,A]	87.1 ± 0.02[bc,B]	83.2 ± 0.02[c]	<0.01	<0.01	<0.01	0.01
Pregnancy by 300 DIM, %	73.2 ± 0.02[a]	69.6 ± 0.02[ab]	62.9 ± 0.03[bc]	56.4 ± 0.03[c]	<0.01	<0.01	<0.01	0.03
Removal from the herd by 300 DIM, %	16.8 ± 0.01[a]	15.5 ± 0.01[a]	23.7 ± 0.02[b]	26.7 ± 0.03[b]	<0.01	<0.01	<0.01	<0.01
Total milk production, kg	10,828 ± 115.6[a]	10,237 ± 157.5[b]	10,036 ± 221.2[bc,A]	9,255 ± 222.4[c,B]	<0.01	<0.01	<0.01	0.45

[a–d]Least squares means with different lowercase superscripts differ (multiple comparison corrected $P \leq 0.05$) within row.

[A,B]Least squares means with different uppercase superscripts tended to differ ($0.05 > P \leq 0.10$) within row (multiple comparison corrected *P*).

[1]NoMet (n = 1,194): cows not diagnosed with metritis; CureD5 (n = 705): cows diagnosed with metritis that achieved clinical cure by 5 d after diagnosis and treatment; CureD12 (n = 406): cows diagnosed with metritis that achieved clinical cure between 6 and 12 d after diagnosis and treatment; NoCure (n = 299): cows diagnosed with metritis that did not achieve clinical cure by 12 d after diagnosis and treatment. Data reported as LSM ± SEM.

[2]Group: overall effect of group (NoMet vs. CureD5 vs. CureD12 vs. NoCure); orthogonal contrasts = Metritis: effect of metritis (NoMet vs. CureD5 + CureD12 + NoCure); Cure: effect of clinical cure (CureD5 + CureD12 vs. NoCure); Time: effect of time to cure (CureD5 vs. CureD12).

within 300 DIM than cows with metritis. Among cows that developed metritis, those that achieved clinical cure had a higher proportion of pregnancy within 300 DIM than those that did not achieve cure. Moreover, the proportion of CureD5 cows that were pregnant by 300 DIM was greater than CureD12 cows. Differences in proportion of pregnant cows by 300 DIM were associated with parity (primiparous = 72.8% vs. multiparous = 58.11%; $P < 0.01$), RFM (No = 60.1% vs. Yes = 70.4%; $P < 0.01$), and calving problem (No = 67.8% vs. Yes = 63.7%; $P = 0.08$).

Differences in hazard of pregnancy within 300 DIM were associated with metritis and clinical cure ($P \leq 0.03$) but not time to cure ($P = 0.43$; Figure 1a). The CuredD5 cows had similar (AHR = 1.06, 95% CI 0.91–1.24) hazard of pregnancy within 300 DIM compared with CuredD12 cows. All pairwise comparisons are available in Supplemental Table S1 (see Notes). In addition, multiparous cows had a lower ($P < 0.01$; AHR = 0.80, 95% CI = 0.72–0.90) hazard of pregnancy within 300 DIM compared with primiparous cows. Cows that did not experience calving problems had a higher ($P = 0.04$; AHR = 1.14, 95% CI 1.01–1.30) hazard of pregnancy within 300 DIM compared with cows that experienced calving problems. Cows not affected by RFM had a higher ($P < 0.01$; AHR = 1.30, 95% CI 1.13–1.51) hazard of pregnancy within 300 DIM compared with cows with RFM. Cows with high BCS at d 0 had a lower ($P = 0.05$; AHR = 0.83, 95% CI 0.70–0.10) hazard of pregnancy within 300 DIM compared with cows with moderate BCS.

Total and Monthly Milk Production and Removal from the Herd

Differences in total milk production within 10 mo of lactation were associated with metritis and cure ($P < 0.01$) but not with time to cure ($P = 0.45$; Table 3). The total milk production was 10,828 ± 115.6 kg for NoMet, 10,237 ± 157.5 kg for CureD5, 10,036 ± 221.2 kg for CureD12, and 9,255 ± 222.4 for NoCure. For cows with metritis, total milk production was reduced compared with cows without metritis. Among cows that developed metritis, those that achieved clinical cure had a greater total milk production than those that did not achieve cure. The CureD5 had similar total milk production compared with CureD12. Moreover, primiparous cows produced less ($P < 0.01$) milk within 10 mo of lactation than multiparous cows (9,381 kg vs 10,796 kg, respectively). A 3-way interaction of group by parity by month of lactation ($P < 0.01$) was associated with differences in monthly milk production for the first 10 mo of lactation. Despite the observed interaction, no differences in monthly milk production were observed between CureD5 and CureD12 cows regardless of parity ($P > 0.10$; Figure 2).

Differences in proportion of removal from the herd by 300 DIM were associated with metritis, cure, and time to cure ($P < 0.01$; Table 3). The proportion of cows that were removed from the herd by 300 DIM was 16.8% ± 0.01% in NoMet, 15.5% ± 0.01% in CureD5, 23.7% ± 0.02% in CureD12, and 26.7% ± 0.03% in NoCure. Cows without metritis had a lower proportion of removal from the herd by 300 DIM compared with cows with metritis. Among cows that developed metritis, those that clinically cured had a lower proportion of removal from

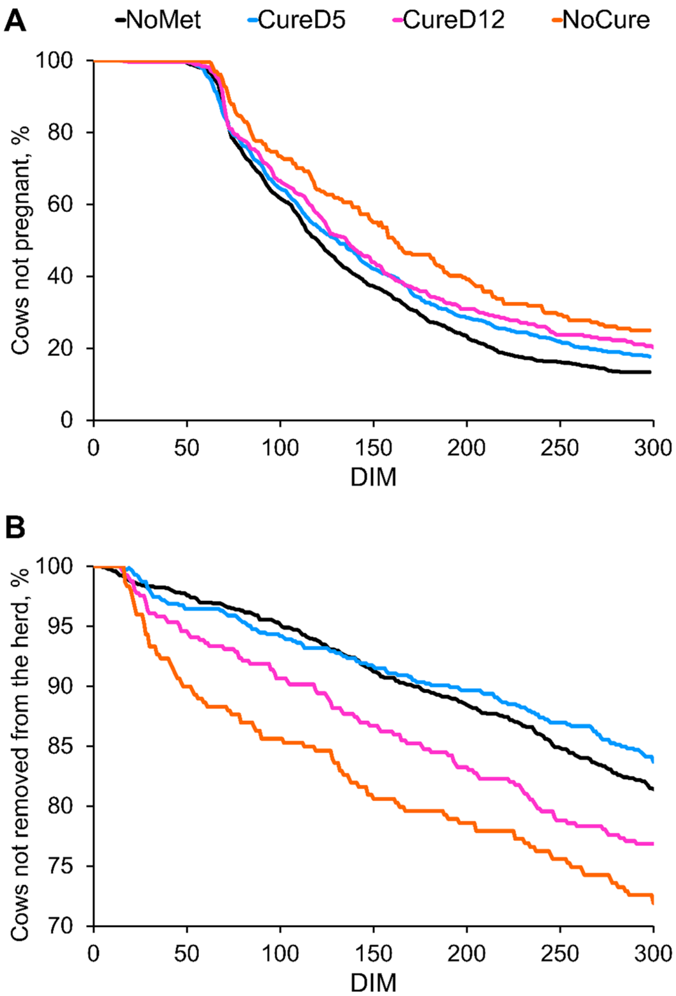

Figure 1. Survival curves for time to pregnancy by 300 DIM (A) for cows not diagnosed with metritis (NoMet; n = 1,194), cows diagnosed with metritis that achieve clinical cure by 5 d after diagnosis and treatment (CureD5; n = 705), cows diagnosed with metritis that achieve clinical cure between 6 and 12 d after diagnosis and treatment (CureD12; n = 406), and cows diagnosed with metritis that did not achieve clinical cure by 12 d after diagnosis and treatment (NoCure; n = 299). Median days to pregnancy: NoMet = 123; CureD5 = 133; CureD12 = 136; NoCure = 163. Percentage of cows censored: NoMet = 12.7; CureD5 = 17.6; CureD12 = 20.6; NoCure = 24.9. Effect of metritis: $P < 0.01$; effect of clinical cure: $P = 0.03$; effect of time to cure: $P = 0.43$. (B) Survival curves for time to removal from the herd by 300 DIM for NoMet, CureD5, CureD12, and NoCure. Percentage of cows censored: NoMet = 81.5; CureD5 = 76.8; CureD12 = 84.1; NoCure = 72.6. Effect of metritis: $P = 0.25$; effect of clinical cure: $P < 0.01$; effect of time to cure: $P < 0.01$.

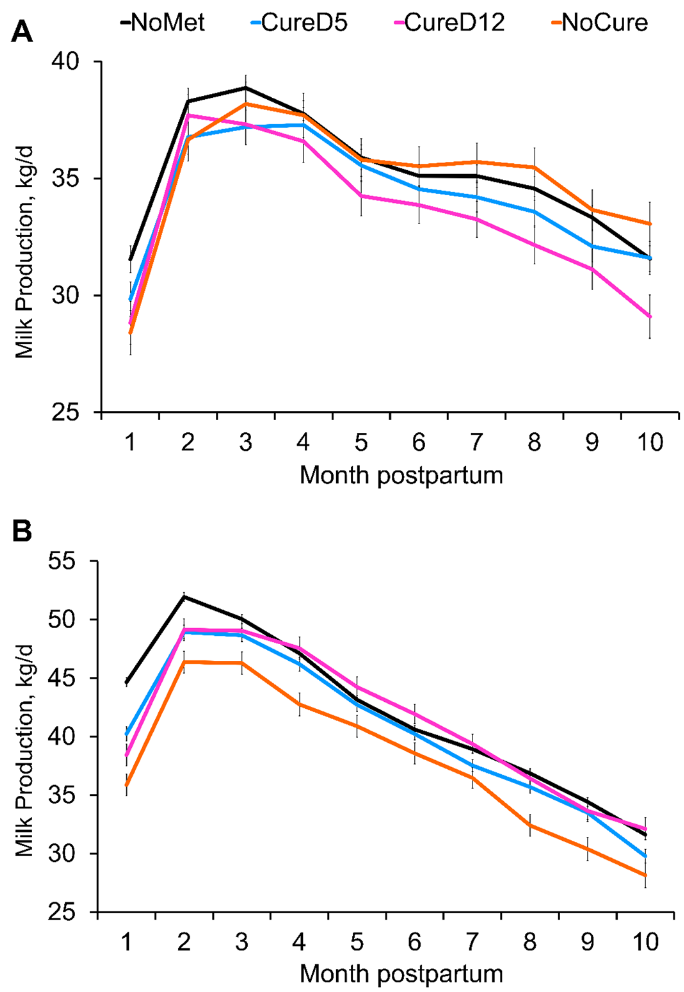

Figure 2. Milk production in the first 10 mo postpartum for primiparous (A) and multiparous cows (B). NoMet (n = 853): cows not diagnosed with metritis; CureD5 (n = 426): cows diagnosed with metritis that achieved clinical cure by 5 d after diagnosis and treatment; CureD12 (n = 212): cows diagnosed with metritis that achieved clinical cure between 6 and 12 d after diagnosis and treatment; NoCure (n = 215): cows diagnosed with metritis that did not achieve clinical cure by 12 d after diagnosis and treatment. Data reported as LSM, and error bars represent SEM. A 3-way interaction of group by parity by month of lactation ($P < 0.01$) was associated with differences in monthly milk production for the first 10 mo of lactation. Despite the observed interaction, no differences in monthly milk production were observed between CureD5 and CureD12 cows regardless of parity ($P > 0.10$).

the herd by 300 DIM compared with cows that did not achieve clinical cure. Moreover, the CureD5 group had a lower proportion of culling compared with the CureD12. Interestingly, no differences ($P > 0.05$) in proportion of removal from the herd by 300 DIM were observed between CureD12 and NoCure. Multiparous cows had a higher proportion of removal from the herd by 300 DIM than primiparous cows (27.0% vs. 15.0%; $P < 0.05$), and cows with RFM had a greater proportion of removal from the herd by 300 DIM than cows without RFM (22.7% vs. 18.1%; $P < 0.05$).

Although hazard of removal from the herd within 300 DIM was not associated with metritis ($P = 0.25$), differences in hazard of removal from the herd within 300 DIM were associated ($P \leq 0.01$) with clinical cure and time to cure (Figure 1b). The CureD5 group had a lower ($P < 0.01$; AHR = 0.61, 95% CI: 0.46–0.80) removal from the herd within 300 DIM than the CureD12 group. All pairwise comparisons are available in Supplemental Table S1. Multiparous cows had a higher ($P < 0.01$; AHR

= 2.06, 95% CI = 1.66–2.56) hazard of removal from the herd within 300 DIM compared with primiparous cows. Cows without RFM tended to have a lower (P = 0.07; AHR = 0.80, 95% CI: 0.63–1.02) hazard of removal from the herd within 300 DIM than that of cows with RFM.

DISCUSSION

This study aimed to assess the association between time to clinical cure and reproductive performance, milk production, and removal from the herd in lactating Holstein cows treated for metritis. Our findings partially supported our hypothesis that cows achieving clinical cure earlier would exhibit better reproductive performance, higher milk production, and reduced risk of removal from the herd than cows that were clinically cured later. Cows that achieved clinical Cure5D of metritis diagnosis had improved reproductive performance, specifically reduced risk of PVD, greater risk of receiving a service, and pregnancy by 300 DIM compared with cows that recovered later. In addition, expedited clinical cure was associated with reduced risk of removal from the herd by 300 DIM compared with those achieving clinical cure of metritis at a later time. Differences in total milk production were not associated with time to clinical cure.

In the present study, multiparous cows had higher odds of achieving CureD5 compared with primiparous cows. Contrasting findings were reported by Menta et al. (2024), in which parity was not associated with different odds of CureD5. In the present dataset, the occurrence of calving problems was similar between primiparous and multiparous cows, suggesting that other factors beyond calving difficulty may explain the association between parity and clinical cure. In contrast, Hiew et al. (2016) reported that dystocic calvings were more frequently observed in primiparous cows compared with multiparous cows. Similarly, a negative correlation between parity and vulvovaginal laceration scoring (**VLS**) was described by Vieira-Neto et al. (2016), with greater VLS observed in primiparous compared with multiparous cows. Furthermore, the occurrence of dystocia and vulvovaginal laceration was previously associated with reduced odds of clinical cure of metritis (Machado et al., 2020), which was also linked to greater blood haptoglobin levels, illustrating a more intense inflammatory process. Cows with RFM or fever on the day of metritis diagnosis had lower odds of CureD5. The occurrence of RFM has been linked to impaired neutrophil function and reduced IL-8 activity, which compromises placental separation and immune defense against uterine pathogens (Kimura et al., 2002; LeBlanc, 2020). In addition, necrotic placental tissue possibly provides a favorable environment for bacterial growth. Fever likely reflects an intensified systemic and uterine inflammatory response as previously described (Machado et al., 2020; Pereira et al., 2025). Inflammation appears to play a central role in clinical cure failure of metritis, as higher concentrations of blood haptoglobin, as well as uterine arachidonic acid and other inflammatory markers, were detected in cows that failed to achieve clinical cure (Machado et al., 2020; Pereira et al., 2025). Together, these findings suggest that systemic inflammation and uterine physical processes, illustrated here as fever and RFM, contribute to delayed clinical recovery. In addition, DIM at metritis diagnosis was positively associated with odds of CureD5 in the present study, indicating that cows diagnosed later postpartum had a greater likelihood of recovery. This finding agrees with previous reports in which DIM was consistently identified as a predictor of cure dynamics. Machado et al. (2020) observed that cows diagnosed after 5 DIM were ~4 times more likely to be CureD12 compared with those diagnosed earlier. Similarly, Menta et al. (2024) reported that each additional DIM at diagnosis increased the odds of cure both at 5 and 14 d after treatment. These consistent findings reinforce the idea that cows diagnosed with metritis earlier postpartum are at greater risk of delayed recovery, likely reflecting the combined challenges during the immediate postpartum period.

Although not formally evaluated herein, a similar unadjusted proportion of cows treated with ceftiofur and ampicillin was observed. The literature regarding the effects of antimicrobial formulation on time to cure is controversial, as differences in time to cure associated with antimicrobial formulation were reported by Lima et al. (2014), but not by Merenda et al. (2021). Herein, we compiled datasets from 2 experiments, incorporating data from the studies by Lima et al. (2014) and Merenda et al. (2021). It is likely that the effects described by Lima et al. (2014) are herd-specific, as only one dairy herd was used in that experiment. A recent study characterized the uterine microbiome of cows with metritis that achieved clinical cure and cows with clinical cure failure of metritis (Figueiredo et al., 2024b). No differences in the uterine microbiome (α- or β-diversity) were associated with clinical cure failure of metritis (Figueiredo et al., 2024b). Moreover, both ceftiofur and ampicillin are β-lactams with broad-spectrum activity, with similar expected effects on the uterine microbiome of cows with metritis (Jeon et al., 2018).

In our study, differences in reproductive performance and risk of removal from the herd were associated with time to clinical cure of metritis in dairy cows. Our findings are partially contrary to those reported by Menta et al. (2024), in which no differences in reproductive performance or risk of removal from the herd associated with time to clinical cure were observed. The discrepancies between the 2 studies likely arise from multiple factors, including the number of enrolled animals and dairy herd locations. For instance, Menta et al. (2024) included 422 cows with metritis from 4 herds located in Texas, California, and Florida, whereas our study included 1,410 cows with metritis from 5 herds located in Florida. Nevertheless, time to clinical cure was associated with different risks of PVD occurrence, which indicates that subsequent uterine health and fertility are partially compromised (Dubuc et al., 2010; Moraes et al., 2021; Ojeda-Rojas et al., 2025). Particularly in our study, the prevalence of PVD between groups was high, contrasting what was observed in previous studies, including multistate surveillance (Pinedo et al., 2020). Such a trend could be illustrating specific conditions related to location of dairies (Florida) and may influence reproductive outcomes presented herein. We also observed an association of expedited clinical cure and the risk of receiving a first service, which may be tied to the similar trend related to PVD. Overall, time to clinical cure was associated with different risks of pregnancy, illustrated by worsened performance for cows that were CureD12 compared with cows that were CureD5. Several studies have depicted the relationship between inflammation, combined uterine diseases, and reproductive performance. For instance, Valdmann et al. (2022) reported that cows affected by endometritis in combination with other calving-related disorders had delayed onset of estrous cyclicity, as well as reduced pregnancy at first service and lower hazard of pregnancy compared with cows affected by endometritis only. Bruinjé et al. (2024b) demonstrated that postpartum dairy cows with high levels of serum haptoglobin (≥0.8 g/L) and endometritis exhibit altered luteal function compared with clinically healthy cows. Wicaksono et al. (2025) also reported the compound effect of multiple reproductive disorders on the lactational performance of dairy cows. The changes in luteal function previously observed (Bruinjé et al., 2019, 2024c) could further explain the impaired reproductive outcomes observed in cows with delayed clinical cure of metritis in our study. Although the mechanisms of subfertility associated with clinical cure of metritis (or time to clinical cure of metritis) have not been completely elucidated, studies indicate that cows with metritis have long-term changes in endometrial morphology, functionality, and inflammation (Sellmer Ramos et al., 2023; Silva et al., 2024; Rashid et al., 2025). It is possible that cows that take longer to achieve clinical cure of metritis present intensified endometrial changes, which reflect on the likelihood of pregnancy within 300 DIM (Ribeiro et al., 2016; Moraes et al., 2021).

The differences in risk of removal from the herd associated with time to cure were expected, especially after considering the uterine health and reproductive performance outcomes previously described herein. A greater proportion of cows that achieved CureD12 were removed from the herd, compared with cows that achieved CureD5. This trend likely occurred due to a combination of factors, including the increased risk of PVD (Dubuc et al., 2010; Moraes et al., 2021; Ojeda-Rojas et al., 2025) and possibly shifts in milk production, which is one of the main reasons for removals from the herd in the United States (USDA, 2018). No differences in milk production were associated with time to clinical cure of metritis (effect size r = 0.07 and Cohen's d = 0.15, depicting small effect size), which indicates that perhaps time to cure of metritis is more closely modulated by a localized inflammatory process in the reproductive tract than a systemic process. It is possible that the similarity in milk production levels between CureD5 and CureD12 cows is at least partially explained by the different proportions of culling. For instance, the greater risk and hazard of culling within 300 DIM in CureD12 cows leads to loss of follow-up, and perhaps differences in milk production would be observed if more CureD12 cows remained in the herd.

Altogether, we observed marked differences in lactational performance associated with time to clinical cure of metritis, particularly in the risk of receiving a service and pregnancy by 300 DIM and the risk of removal from the herd. Risk factors for time to clinical cure of metritis were linked to physical injury and inflammatory processes, rather than bacterial communities, which suggests that extralabel antimicrobial use for metritis is not beneficial. It is important to highlight that the present study is comprised of retrospective data generated by 3 experiments conducted in different dairy herds in Florida. Despite the significant number of animals per group in our study, a formal sample size calculation was not performed; furthermore, all cows were located in a single state, which may limit how translatable the outcomes observed herein are to cows located in different regions and climates. Moreover, it is possible that outcomes assessed by 300 DIM may obscure early negative effects of metritis and clinical cure failure. More studies are warranted to assess the effects of time to cure and to develop strategies for managing cows that fail to achieve clinical cure, as it seems that lactational performance does not equate to that observed in clinically cured cows, even after antimicrobial use.

CONCLUSIONS

Cows that achieved clinical Cure5D after antimicrobial therapy had improved reproductive performance (risk of receiving a first service and pregnancy by 300 DIM) and a lower risk of removal from the herd compared with those that recovered later. Time to clinical cure of metritis was not associated with differences in total milk production. Time to cure of metritis may be a useful indicator of future performance and survivability in dairy herds and may help guide on-farm decisions and identify cows at greater risk for suboptimal outcomes.

NOTES

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Nonstandard abbreviations used: AI = artificial insemination; CureD5 = clinical cure by d 5; CureD12 = clinical cure between d 6 and 12; NoCure = did not achieve clinical cure by d 12; NoMet = cows without metritis; OR = odds ratios; PVD = purulent VD; RFM = retained fetal membranes; RT = rectal temperature; VD = vaginal discharge; VIF = variance inflation factor; VLS = vulvovaginal laceration scoring; VWP = voluntary waiting period.

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